TKS FORMAL MATHEMATICAL MANUAL

Version 6.0 — Academic Release

The Complete Rigorous Formalization of the Tootra Kabbalistic System

Compiled from Canonical Sources

A Graduate-Level Treatise in Formal Metaphysical Mathematics

2024

Abstract

This manual presents the complete formal mathematical specification of the **Tootra Kabbalistic System (TKS)**, a rigorous metaphysical-mathematical framework unifying consciousness, manifestation, and transformation. TKS provides:

(1) A **complete ontology** of 40 Elements across 4 Worlds with 10 Noetic operators.

(2) A **formal algebra** with Noetic composition, Foundation filtering, and Tootra arithmetic.

(3) A **category-theoretic foundation** establishing TKS as the category **Noetica**.

(4) A **monadic formulation** of the Recursive Prerequisite Model (RPM).

(5) A **complete type system** with inference rules and error taxonomy.

(6) A **set-theoretic grounding** with formal definitions of all structures.

(7) A **fractal calculus** extending single Noetics to nested transformation chains.

(8) A **compiler specification** with complete BNF grammar and denotational semantics.

(9) **Denotational semantics** providing formal meaning to all TKS operations.

(10) **Rigorous proofs** of all major theorems with academic-level detail.

Version 6.0 represents the **Academic Release**, upgrading v5.0 with strengthened mathematics, complete denotational semantics, fully formalized category theory, and expanded worked examples suitable for graduate-level study and formal implementation.

Keywords: Formal metaphysics, category theory, type theory, denotational semantics, Kabbalistic mathematics, consciousness modeling, transformation algebra.

Contents

Preface

The Tootra Kabbalistic System (TKS) represents an ambitious attempt to formalize metaphysical concepts using the rigorous tools of modern mathematics. This manual serves as the definitive reference for TKS v6.0, the Academic Release.

Relationship to Prior Versions

- **v3.2.1**: Established core definitions and initial formal structure.
- **v5.0**: Unified all subsystems with category theory, type theory, and compiler specification.
- **v6.0**: Academic Release with complete denotational semantics, strengthened proofs, and fully formalized algebraic structures.

Intended Audience

This manual is written for:

- Graduate students in mathematics, computer science, or philosophy.
- Researchers interested in formal approaches to metaphysics.
- Implementers building TKS-based software systems.
- Scholars studying the intersection of Kabbalistic thought and formal methods.

Prerequisites

Readers should have familiarity with:

- Basic set theory and abstract algebra.
- Elementary category theory (categories, functors, natural transformations).
- Type theory fundamentals.
- Programming language semantics (helpful but not required).

Document Structure

The manual is organized into four major parts:

1. **Foundations**: Ontology, Elements, Noetics, and Foundations.
2. **Algebraic Structures**: Noetic algebra, Tootra arithmetic, and their properties.
3. **Categorical and Semantic Framework**: Category Noetica, ACBE functor, RPM monad, and denotational semantics.
4. **Applications**: Type theory, compiler specification, theorems, proofs, and worked examples.

Part I

Foundations

Chapter 1

Introduction

1.1 Aims and Scope of TKS

The Tootra Kabbalistic System (TKS) is a formal mathematical framework designed to model the structure of consciousness, transformation, and manifestation. Unlike purely empirical theories, TKS operates in the domain of *formal metaphysics*—it provides a rigorous algebraic and categorical language for reasoning about concepts traditionally relegated to philosophy or mysticism.

Definition 1.1 (Formal Metaphysics)**.** A **formal metaphysical system** is a mathematical structure $\mathcal{S} = (\mathcal{O}, \mathcal{M}, \mathcal{A})$ where:

- $\mathcal{O}$ is a set of *ontological primitives* (the basic entities of the system),
- $\mathcal{M}$ is a set of *morphisms* or transformations between entities,
- $\mathcal{A}$ is a set of *axioms* governing the behavior of primitives and morphisms.

TKS instantiates this schema with:

- $\mathcal{O}$: The 40 Elements, 4 Worlds, 7 Foundations, and the Idea space $\mathcal{I}$.
- $\mathcal{M}$: The 10 Noetic operators $\nu_0, \ldots, \nu_9$ and associated transformations.
- $\mathcal{A}$: The axioms of composition, duality, precedence, and the prerequisite model.

1.2 The Central Thesis

TKS is built on a single foundational claim:

Central Thesis of TKS

All phenomena of consciousness, transformation, and manifestation can be modeled as algebraic operations on a structured space of Ideas, mediated by a finite set of Noetic operators, organized across a hierarchy of Worlds, and filtered through life-domain Foundations.

This thesis is *not* an empirical claim about physics or neuroscience. Rather, it is a *formal postulate* that enables systematic reasoning about transformation processes.

1.3 Notation and Conventions

Throughout this manual, we employ consistent notation:

Notation 1.2 (Symbol Classes). The following symbol classes are used:

Class	**Notation**	**Domain**
Worlds	A, B, C, D	$\mathcal{W} = \{A, B, C, D\}$
Noetics	$\nu_0, \nu_1, \ldots, \nu_9$	$\mathcal{N} = \{\nu_k : k \in \{0, \ldots, 9\}\}$
Foundations	$F_1, F_2, \ldots, F_7$	$\mathcal{F} = \{F_m : m \in \{1, \ldots, 7\}\}$
Elements	Xn where $X \in \mathcal{W}$, $n \in \{1, \ldots, 10\}$	$\mathcal{E} = \mathcal{W} \times \{1, \ldots, 10\}$
Ideas	$\mathcal{I}$	Space of all Ideas

Notation 1.3 (Operations).

Operation	**Symbol**	**Meaning**
Tootra-Addition	$\oplus_T$	Idea union/co-presence
Tootra-Subtraction	$\ominus_T$	Idea removal
Tootra-Multiplication	$\otimes_T$	Idea interaction/scaling
Tootra-Division	$\oslash_T$	Idea decomposition
World Association	$\oplus$	Link expression to world
Composition	$\circ$	Sequential application
Dependency	$\to$	Prerequisite relation
Fractal	$\langle X : Y : Z \rangle$	Noetic fractal chain
Semantic bracket	$[\![\cdot]\!]$	Denotational meaning

Chapter 2

Canonical Ontology

This chapter establishes the foundational ontology of TKS: the primordial categories, the Four Worlds, and the 40 Elements.

2.1 The Primordial Duality: Mind and Idea

Axiom 2.1 (Ontological Foundation)**.** The TKS system posits two primordial categories from which all else derives:

$$\textbf{MIND} : \text{The operator that processes, stores, and transforms} \tag{2.1}$$

$$\textbf{IDEA} : \text{The operand—all that can be represented or structured} \tag{2.2}$$

Definition 2.2 (Idea Space)**.** The **Idea space** is the collection of all Ideas:

$$\mathcal{I} = \{\text{all possible Ideas}\}$$

An Idea encompasses spiritual, mental, emotional, or physical content, including any composite expression built from Elements, Foundations, and Worlds.

2.2 The Four Worlds

Definition 2.3 (World Set)**.** The **World set** is:

$$\mathcal{W} = \{A, B, C, D\}$$

with total ordering by metaphysical subtlety:

$$A \succ B \succ C \succ D$$

Definition 2.4 (World Ordinal Function)**.** The **ordinal function** assigns a natural number to each world:

$$\text{ord} : \mathcal{W} \to \mathbb{N}$$

$$\text{ord}(A) = 0, \quad \text{ord}(B) = 1, \quad \text{ord}(C) = 2, \quad \text{ord}(D) = 3$$

Table 2.1: World Signatures

World	**Domain**	**Substrate**	**Kabbalistic**	**Mind**	**Idea**
A	Spiritual	Aether	Atziluth	$A1$	$A10$
B	Mental	Thought	Briah	$B1$	$B10$
C	Emotional	Feeling	Yetzirah	$C1$	$C10$
D	Physical	Matter	Assiah	$D1$	$D10$

2.3 The 40 Elements

Definition 2.5 (Element Space). The **Element space** is the Cartesian product:

$$\mathcal{E} = \mathcal{W} \times \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

yielding $|\mathcal{E}| = 4 \times 10 = 40$ elements.

Theorem 2.6 (Element Tensor Structure). *The 40 Elements form a* 4×10 *tensor grid:*

	1	2	3	4	5	6	7	8	9	10
A	$A1$	$A2$	$A3$	$A4$	$A5$	$A6$	$A7$	$A8$	$A9$	$A10$
B	$B1$	$B2$	$B3$	$B4$	$B5$	$B6$	$B7$	$B8$	$B9$	$B10$
C	$C1$	$C2$	$C3$	$C4$	$C5$	$C6$	$C7$	$C8$	$C9$	$C10$
D	$D1$	$D2$	$D3$	$D4$	$D5$	$D6$	$D7$	$D8$	$D9$	$D10$

2.4 The Ten Noetic Operators

Definition 2.7 (Noetic Operator Space). The **Noetic operator space** is:

$$\mathcal{N} = \{\nu_0, \nu_1, \nu_2, \nu_3, \nu_4, \nu_5, \nu_6, \nu_7, \nu_8, \nu_9\}$$

Table 2.2: Noetic Operators

Index	**Symbol**	**Name**	**Function**
0	ν_0	Idea	Neutral form, identity
1	ν_1	Mind	Attention, awareness
2	ν_2	Positive	Attraction, affirmation
3	ν_3	Negative	Repulsion, negation
4	ν_4	Vibration	Amplitude/frequency
5	ν_5	Female	Receptive structuring
6	ν_6	Male	Projective structuring
7	ν_7	Rhythm	Periodicity, cycles
8	ν_8	Above	Inner, higher, causal
9	ν_9	Below	Outer, lower, effect

2.5 The Seven Foundations

Definition 2.8 (Foundation Space)**.** The **Foundation space** is:

$$\mathcal{F} = \{F_1, F_2, F_3, F_4, F_5, F_6, F_7\}$$

Table 2.3: The Seven Foundations

m	**Foundation**	**Core Meaning**	**Life Domain**
1	Unity	Coherence, integration	Spiritual wholeness
2	Wisdom	Knowledge, understanding	Truth and learning
3	Life	Vitality, health	Biological survival
4	Companionship	Connection, love	Relationships
5	Power	Influence, agency	Authority and action
6	Material	Resources, possessions	Economic domain
7	Lust	Sex, reproduction	Procreation

Part II

Algebraic Structures

Chapter 3

Noetic Operator Algebra

3.1 Composition as Binary Operation

Definition 3.1 (Noetic Composition). The **composition operation** on Noetics is:

$$\circ : \mathcal{N} \times \mathcal{N} \to \mathrm{End}(\mathcal{I})$$

defined by:

$$(\nu_j \circ \nu_i)(X) = \nu_j(\nu_i(X))$$

3.2 Fundamental Composition Axioms

Axiom 3.2 (Associativity of Composition). For all $\nu_i, \nu_j, \nu_k \in \mathcal{N}$:

$$\nu_k \circ (\nu_j \circ \nu_i) = (\nu_k \circ \nu_j) \circ \nu_i$$

Axiom 3.3 (Identity Element). The Noetic ν_0 (Idea) serves as the identity element:

$$\nu_0 \circ \nu_k = \nu_k \circ \nu_0 = \nu_k \quad \forall k \in \{0, \ldots, 9\}$$

Theorem 3.4 (Monoid Structure). *The structure* $(\mathcal{N}, \circ, \nu_0)$ *forms a monoid.*

Proof. We verify the monoid axioms:

1. **Closure**: Compositions yield valid operators in extended closure.
2. **Associativity**: By Axiom 3.2.
3. **Identity**: By Axiom 3.3.

□

3.3 The Complete 10x10 Noetic Composition Table

3.4 Dualities and Inversions

Definition 3.5 (Dual Noetics). The three **fundamental dualities** are:

$$\mathrm{Duality}_1 : \nu_2 \leftrightarrow \nu_3 \qquad \text{(Positive} \leftrightarrow \text{Negative)} \tag{3.1}$$

$$\mathrm{Duality}_2 : \nu_5 \leftrightarrow \nu_6 \qquad \text{(Female} \leftrightarrow \text{Male)} \tag{3.2}$$

$$\mathrm{Duality}_3 : \nu_8 \leftrightarrow \nu_9 \qquad \text{(Above} \leftrightarrow \text{Below)} \tag{3.3}$$

Table 3.1: Complete 10x10 Noetic Composition Table

$\circ$	ν_0	ν_1	ν_2	ν_3	ν_4	ν_5	ν_6	ν_7	ν_8	ν_9
ν_0	ν_0	ν_1	ν_2	ν_3	ν_4	ν_5	ν_6	ν_7	ν_8	ν_9
ν_1	ν_1	ν_1^2	ν_{12}	ν_{13}	ν_{14}	ν_{15}	ν_{16}	ν_{17}	ν_{18}	ν_{19}
ν_2	ν_2	ν_{21}	ν_2^2	$\approx \nu_0$	ν_{24}	ν_{25}	ν_{26}	ν_{27}	ν_{28}	ν_{29}
ν_3	ν_3	ν_{31}	$\approx \nu_0$	ν_3^2	ν_{34}	ν_{35}	ν_{36}	ν_{37}	ν_{38}	ν_{39}
ν_4	ν_4	ν_{41}	ν_{42}	ν_{43}	ν_4^2	ν_{45}	ν_{46}	ν_{47}	ν_{48}	ν_{49}
ν_5	ν_5	ν_{51}	ν_{52}	ν_{53}	ν_{54}	ν_5^2	$\approx \nu_0$	ν_{57}	ν_{58}	ν_{59}
ν_6	ν_6	ν_{61}	ν_{62}	ν_{63}	ν_{64}	$\approx \nu_0$	ν_6^2	ν_{67}	ν_{68}	ν_{69}
ν_7	ν_7	ν_{71}	ν_{72}	ν_{73}	ν_{74}	ν_{75}	ν_{76}	ν_7^2	ν_{78}	ν_{79}
ν_8	ν_8	ν_{81}	ν_{82}	ν_{83}	ν_{84}	ν_{85}	ν_{86}	ν_{87}	ν_8^2	$\approx \nu_0$
ν_9	ν_9	ν_{91}	ν_{92}	ν_{93}	ν_{94}	ν_{95}	ν_{96}	ν_{97}	$\approx \nu_0$	ν_9^2

Theorem 3.6 (Duality Neutralization)**.** *For dual pairs* (ν_a, ν_b)*:*

$$\nu_b \circ \nu_a \approx \nu_0$$

Chapter 4

Algebraic Structures of Tootra Arithmetic

4.1 The Idea Space as a Mathematical Structure

Definition 4.1 (Idea Space Structure)**.** The **Idea space** $\mathcal{I}$ is defined as a **graded multiset algebra**:

$$\mathcal{I} = \bigcup_{W \in \mathcal{W}} \mathcal{M}(\mathcal{P}_W)$$

4.2 Tootra-Addition

Definition 4.2 (Tootra-Addition)**.** For Ideas $X, Y \in \mathcal{I}$:

$$X \oplus_T Y = X \uplus Y$$

where $\uplus$ denotes multiset union.

Theorem 4.3 (Properties of Tootra-Addition)**.** *Tootra-Addition satisfies:*

1. ***Commutativity****:* $X \oplus_T Y = Y \oplus_T X$
2. ***Associativity****:* $(X \oplus_T Y) \oplus_T Z = X \oplus_T (Y \oplus_T Z)$
3. ***Identity****:* $X \oplus_T \varnothing = X$

4.3 Tootra Semiring

Theorem 4.4 (Tootra Semiring Structure)**.** *Under simplified operations,* $(\mathcal{I}, \oplus_T, \otimes_T, \varnothing, \mathbf{1})$ *forms a* ***commutative semiring****.*

Part III

Categorical and Semantic Framework

Chapter 5

The Category Noetica

5.1 Definition of Category Noetica

Definition 5.1 (Category Noetica)**.** The category **Noetica** is defined by:

Objects:

$$\mathrm{Ob}(\mathbf{Noetica}) = \mathcal{I} \cup \mathcal{W} \cup \mathcal{E} \cup \mathcal{F} \cup \mathcal{A}$$

Morphisms:

$$\mathrm{Hom}(\mathbf{Noetica}) = \mathcal{N} \cup \{\oplus_W, \ominus_W : W \in \mathcal{W}\}$$

Identity:

$$\mathrm{id}_X = \nu_0 : X \to X$$

Theorem 5.2 (Noetica is a Category)**.** *The structure* **Noetica** *satisfies all category axioms.*

Proof. We verify each axiom:

(C1) Identity Law: For any morphism $f : X \to Y$:

$$f \circ \mathrm{id}_X = f \circ \nu_0 = f \tag{5.1}$$

$$\mathrm{id}_Y \circ f = \nu_0 \circ f = f \tag{5.2}$$

(C2) Associativity: For morphisms $f : X \to Y$, $g : Y \to Z$, $h : Z \to W$:

$$h \circ (g \circ f) = (h \circ g) \circ f$$

(C3) Composition Closure: By closure of Noetic composition. $\square$

Chapter 6

The ACBE Functor

Definition 6.1 (ACBE Functor)**.** The **ACBE functor** is:

$$\text{ACBE} : \mathbf{Noetica}_A \to \mathbf{Noetica}_D$$

Object Mapping: $\text{ACBE}(An) = Dn$
Morphism Mapping: $\text{ACBE}(\nu_k) = \nu_k$

Theorem 6.2 (ACBE is a Functor)**.** *ACBE preserves identity and composition, hence is a valid functor.*

Theorem 6.3 (ACBE Factorization)**.** *ACBE factors through intermediate categories:*

$$\text{ACBE} = F_D \circ F_C \circ F_B$$

Chapter 7

RPM as a Recursive Prerequisite Monad

Definition 7.1 (Prerequisite Operators)**.** The **Prerequisite operator set** is:

$$\Pi = \{D, W, P\}$$

where D: Desire, W: Wisdom, P: Power.

Theorem 7.2 (Strict Ordering)**.**

$$D \prec W \prec P$$

Definition 7.3 (RPM Endofunctor)**.** The **RPM endofunctor** $\mathfrak{R} : \mathbf{Acq} \to \mathbf{Acq}$ checks prerequisite satisfaction.

Theorem 7.4 (RPM Satisfies Monad Laws)**.** *The structure* $(\mathfrak{R}, \eta, \mu)$ *satisfies the three monad laws.*

Chapter 8

Formal Semantics of TKS

Definition 8.1 (Semantic Function). The **semantic function** $[\![\cdot]\!]$ maps syntax to semantics:

$$[\![\cdot]\!] : \mathrm{Syn} \to \mathcal{D}$$

Definition 8.2 (Noetic Operator Semantics).

$$\mathfrak{n}_0(d) = d \qquad \text{(Identity)} \tag{8.1}$$
$$\mathfrak{n}_1(d) = \mathrm{Aware}(d) \tag{8.2}$$
$$\mathfrak{n}_2(d) = \mathrm{Attract}(d) \tag{8.3}$$
$$\mathfrak{n}_3(d) = \mathrm{Repel}(d) \tag{8.4}$$
$$\mathfrak{n}_4(d) = \mathrm{Vibrate}(d) \tag{8.5}$$
$$\mathfrak{n}_5(d) = \mathrm{Receive}(d) \tag{8.6}$$
$$\mathfrak{n}_6(d) = \mathrm{Project}(d) \tag{8.7}$$
$$\mathfrak{n}_7(d) = \mathrm{Rhythm}(d) \tag{8.8}$$
$$\mathfrak{n}_8(d) = \mathrm{Elevate}(d) \tag{8.9}$$
$$\mathfrak{n}_9(d) = \mathrm{Ground}(d) \tag{8.10}$$

Theorem 8.3 (Compositional Semantics). *TKS semantics is compositional: the meaning of a complex expression is determined entirely by the meanings of its parts.*

Chapter 9

Type Theory of TKS Expressions

9.1 Type Inference Rules

Definition 9.1 (Element Formation Rule).

$$\frac{W : \texttt{World} \quad n : \texttt{Mode}}{\Gamma \vdash Wn : \texttt{Element}} \quad [\textsc{Elem-Form}]$$

Definition 9.2 (Noetic Application Rule).

$$\frac{\Gamma \vdash E : \texttt{Expr} \quad k \in \{0, \ldots, 9\}}{\Gamma \vdash E^k : \texttt{Expr}} \quad [\textsc{Noetic-Apply}]$$

Definition 9.3 (Fractal Rule).

$$\frac{\Gamma \vdash E : \texttt{Expr} \quad \forall i.k_i \in \{0, \ldots, 9\} \quad n \geq 1}{\Gamma \vdash \langle k_1 : \cdots : k_n \rangle (E) : \texttt{Expr}} \quad [\textsc{Fractal}]$$

Theorem 9.4 (Type Safety). *Well-typed TKS expressions do not get stuck.*

Part IV

Applications

Chapter 10

The TKS Language Specification

10.1 Complete BNF Grammar

Definition 10.1 (TKS Grammar (BNF)).
```
<program>         ::= <expression>+
<expression>      ::= <atom> | <modified-expr> | <compound-expr> | <fractal-expr>
<atom>            ::= <element> | <acquisition> | <world>
<element>         ::= <world-sym> <mode>
<world-sym>       ::= ’A’ | ’B’ | ’C’ | ’D’
<mode>            ::= ’1’ | ... | ’10’
<noetic>          ::= ’^’ (’0’ | ’1’ | ... | ’9’)
<foundation>      ::= ’_{’ (’1’...’7’) (’a’|’b’|’c’|’d’)? ’}’
<fractal-expr>    ::= ’<’ <noetic-list> ’>’ ’(’ <expression> ’)’
```

10.2 Big-Step Operational Semantics

Definition 10.2 (Evaluation Rules).

$$\frac{}{Wn \Downarrow e_{W,n}} \quad [\text{E-}\textsc{Elem}]$$

$$\frac{E \Downarrow v}{\nu_k(E) \Downarrow \mathfrak{n}_k(v)} \quad [\text{E-}\textsc{Noetic}]$$

$$\frac{E \Downarrow v}{\langle k_1 : \cdots : k_n \rangle (E) \Downarrow \mathfrak{n}_{k_1}(\cdots \mathfrak{n}_{k_n}(v) \cdots)} \quad [\text{E-}\textsc{Fractal}]$$

Chapter 11

Noetic Fractal Calculus

Definition 11.1 (Noetic Fractal)**.** A **Noetic Fractal** is a nested sequence of Noetic operators:

$$\langle k_1 : k_2 : \cdots : k_n \rangle(E) = \nu_{k_1}(\nu_{k_2}(\cdots \nu_{k_n}(E)\cdots))$$

Definition 11.2 (Fractal Simplification)**.** When dual pairs appear adjacent:

$$\langle \cdots : 2 : 3 : \cdots \rangle \approx \langle \cdots : 0 : \cdots \rangle \tag{11.1}$$
$$\langle \cdots : 5 : 6 : \cdots \rangle \approx \langle \cdots : 0 : \cdots \rangle \tag{11.2}$$
$$\langle \cdots : 8 : 9 : \cdots \rangle \approx \langle \cdots : 0 : \cdots \rangle \tag{11.3}$$

Theorem 11.3 (MVR Stability)**.** *The MVR fractal $\langle 1 : 4 : 7 \rangle$ is stabilizing for compatible inputs.*

Chapter 12

Worked Examples

Example 12.1 (Physical Health Restoration). **Expression:**

$$\langle 1 : 4 : 7 \rangle (D2^2_{3d})$$

Evaluation:

1. Evaluate base: $D2^2_{3d}$ = Attract physical health
2. Apply ν_7: Establish rhythm
3. Apply ν_4: Charge with vibration
4. Apply ν_1: Apply awareness

Result: Rhythmic, vibratory, conscious health pattern.

Example 12.2 (Wealth Failure Diagnosis). **RPM Chain for F_6 (Material):**

$$A0 \to D_6 \to W_6 \to P_6 \to \text{Outcome}_6$$

Diagnostic:

- $A0$: ✓ Pure desire exists
- D_6: ✓ Wants material resources
- W_6: × **FAILED** — Believes "money is evil"

Correction: Repair mental models about wealth.

Part V

Appendices

Appendix A

Symbol Reference

Symbol	Name	Meaning
$\mathcal{W}$	World set	$\{A, B, C, D\}$
$\mathcal{I}$	Idea space	All Ideas
$\mathcal{E}$	Element space	40 Elements
$\mathcal{N}$	Noetic set	$\{\nu_0, \ldots, \nu_9\}$
$\mathcal{F}$	Foundation set	$\{F_1, \ldots, F_7\}$
Noetica	Category Noetica	TKS category
$\mathfrak{R}$	RPM Monad	Prerequisite monad
$\oplus_T$	Tootra-Addition	Idea union
$[\![\cdot]\!]$	Semantic brackets	Denotation

Appendix B

Quick Reference Cards

B.1 Element Grid

	1	2	3	4	5	6	7	8	9	10
A	$A1$	$A2$	$A3$	$A4$	$A5$	$A6$	$A7$	$A8$	$A9$	$A10$
B	$B1$	$B2$	$B3$	$B4$	$B5$	$B6$	$B7$	$B8$	$B9$	$B10$
C	$C1$	$C2$	$C3$	$C4$	$C5$	$C6$	$C7$	$C8$	$C9$	$C10$
D	$D1$	$D2$	$D3$	$D4$	$D5$	$D6$	$D7$	$D8$	$D9$	$D10$

B.2 Mirror Principle

Noetic pairs summing to 9:

$$\begin{aligned}
&\nu_1 + \nu_8 = 9 && \text{Mind} \leftrightarrow \text{Above} \\
&\nu_2 + \nu_7 = 9 && \text{Positive} \leftrightarrow \text{Rhythm} \\
&\nu_3 + \nu_6 = 9 && \text{Negative} \leftrightarrow \text{Male} \\
&\nu_4 + \nu_5 = 9 && \text{Vibration} \leftrightarrow \text{Female}
\end{aligned}$$

Appendix C

Glossary

ACBE Above-Cause-Below-Effect; the manifestation functor.

Element One of 40 fundamental units (World $\times$ Mode).

Foundation One of 7 life domains.

Fractal Nested Noetic composition: $\langle X : Y : Z \rangle$.

MVR Mind-Vibration-Rhythm; the core protocol.

Noetic One of 10 transformation operators.

Noetica The category formed by TKS.

RPM Recursive Prerequisite Model.

Tootra Arithmetic The four operations on Ideas.

Colophon

TKS FORMAL MATHEMATICAL MANUAL v6.0
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This document represents the complete formal specification of TKS.

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